

PHYSICAL CHEMISTRY

(CHEM-313)

(PAST PAPER-2020)

SHORT QUESTIONS

Q#1

WHAT IS CLOSED PROCESS?

A closed process refers to a system or a process in which no matter enters or leaves the system. It is a controlled system where the total amount of matter within the system remains constant throughout the process.

Q#2

WHAT IS PARTITION FUNCTION?

The partition function is a concept used in statistical mechanics and thermodynamics to describe the distribution of energy among the different energy states of a system.

In a simplified explanation, the partition function provides information about the number of ways a system can distribute its energy among its available energy levels.

Q#3

DEFINE SECOND LAW OF THERMODYNAMICS?

The second law can be stated in various forms, but a common definition is:

"The entropy of an isolated system tends to increase over time, and natural processes are characterized by an increase in the total entropy of the system and its surroundings."

Q#4

WHAT IS MEANT BY DEGREE OF FREEDOM?

In the context of thermodynamics and statistical mechanics, the term "degree of freedom" refers to the independent ways in which a system can store or distribute energy. It represents the number of parameters or variables that need to be specified to completely describe the state of a system.

Q#5

DESCRIBE MOST PROBABLE VELOCITY?

The most probable velocity, also known as the mode of the velocity distribution, refers to the velocity at which the highest number of particles in a gas or fluid have that particular velocity. It is a concept used in the study of kinetic theory and statistical mechanics to describe the velocity distribution of particles in a system.

Q#6

WHAT IS CAGE EFFECT?

The cage effect, also known as the solvent cage effect or solvent shell effect, refers to a phenomenon observed in chemical reactions where the solvent molecules surrounding a reactant or intermediate species can influence the reaction dynamics and kinetics.

When a reactant or intermediate species is surrounded by solvent molecules, they form a "cage-like" structure around the molecule, creating a transient solvent shell.

Q#7

REACTIONS IN SOLUTION PHASE ARE GENERALLY FASTER THAN IN GAS PHASE?

In general, reactions in the solution phase tend to be faster than in the gas phase. This is primarily due to the following factors:

a) Increased molecular collisions: In a solution, the molecules are closely packed and have higher chances of colliding with each other compared to the gas phase, where molecules are more spread out.

b) Enhanced molecular mobility: In a solution, molecules have greater freedom of movement due to the liquid nature of the solvent. This enhanced mobility allows reactant molecules to explore a larger volume and increases the likelihood of encountering other reactant molecules, thereby promoting faster reaction rates.

Q#8

WHAT IS LAW OF CONSERVATION OF ENERGY?

The law of conservation of energy, also known as the first law of thermodynamics or the principle of energy

conservation, states that energy cannot be created or destroyed in an isolated system. It can only be transformed from one form to another or transferred between different objects or systems.

Q#9

WHAT IS DIFFERENCE BETWEEN THE SPONTANEOUS AND NON-SPONTANEOUS REACTION?

SPONTANEOUS REACTION VERSUS NONSPONTANEOUS REACTION	
Spontaneous reactions refer to the chemical reactions that occur without being driven by an outside force	Nonspontaneous reactions refer to the chemical reactions that require an energy input to proceed
Favourable	Unfavourable
Do not require energy input to proceed under a set of conditions	Require an energy input to proceed
The change in the Gibbs free energy is negative	The change in the Gibbs free energy is positive
Release the free energy from the system, making it more stable	Increase the total energy of the system
The total energy of the products is less than that of the reactants	The total energy of the products is higher than that of the reactants
Have a negative enthalpy	Have a positive enthalpy

Q#10

EXPLAIN THE NERST HEAT THEORM?

The Nernst heat theorem says that as absolute zero is approached, the entropy change ΔS for a chemical or physical transformation approaches 0. This can be expressed mathematically as follows:

$$\lim_{T \rightarrow 0} \Delta S = 0$$

(LONG QUESTIONS)

Q#1

**DISCUSS THIRD LAW OF
THERMODYNAMICS AND GIVE ITS
EXPERIMENTAL VERIFICATION?**

**(FOR ANSWER SEE MODERN PHYSICAL CHEMISTRY
VOLUME # 1 BY CARAVAN PAGE NO 236-239).**

Q#2

**DERIVE AND EXPLAIN MAXWELL
DISTRIBUTION LAW FOR DISTRIBUTIONS OF
MOLECULAR VELOCITIES AMONG GAS
MOLECULES?**

**(FOR ANSWER SEE MODERN PHYSICAL CHEMISTRY
VOLUME # 1 BY CARAVAN PAGE NO 15-20).**

Q#3

(A) DISCUSS EFFECT OF TEMPERATURE ON RATE OF REACTION ON THE BASIS OF ARRHENIOUS EQUATION?

The Arrhenius equation provides a quantitative relationship between the rate of a chemical reaction and temperature. It is expressed as:

$$k = A * \exp(-E_a/RT)$$

where:-

k is the rate constant of the reaction,

A is the pre-exponential factor or frequency factor,

E_a is the activation energy,

R is the gas constant (8.314 J/(mol·K)),

T is the absolute temperature in Kelvin.

The Arrhenius equation suggests that the rate constant (k) of a reaction increases exponentially with an increase in temperature (T). This relationship can be understood by considering the following effects of temperature on the rate of reaction:

Activation energy: *The activation energy (E_a) represents the minimum energy required for a reaction to occur. As the temperature increases, the average kinetic energy of the reactant molecules also increases.*

Collision frequency: *Temperature affects the collision frequency of reactant molecules. As temperature rises, the*

kinetic energy and speed of the molecules increase, leading to more frequent collisions between reactant molecules.

Reaction rate constant: *The rate constant (k) in the Arrhenius equation represents the proportionality between the rate of reaction and the concentrations of reactants. The exponential term, $\exp(-E_a/RT)$, takes into account the temperature dependence of the rate constant.*

Temperature dependence: *The Arrhenius equation highlights the temperature dependence of the reaction rate. A small change in temperature can have a significant effect on the rate constant, leading to a substantial change in the reaction rate.*

- In summary, the Arrhenius equation demonstrates that an increase in temperature enhances the reaction rate by increasing the proportion of reactant molecules with sufficient energy to overcome the activation energy barrier.*

(B) DERIVE EYRING EQUATION USING POSTULATES OF TRANSITION STATE THEORY?

The Eyring equation, also known as the Eyring-Polanyi equation, is derived from transition state theory and provides a mathematical expression for the rate constant of a chemical reaction. The Eyring equation is given as:

$$k = (k_b * T / h) * e^{(-\Delta G^\ddagger / RT)}$$

where:-

- *k* is the rate constant of the reaction.
- *k_b* is the Boltzmann constant.
- *T* is the temperature in Kelvin.
- *h* is the Planck constant.
- $\Delta G^\ddagger$ is the Gibbs free energy of activation for the reaction's transition state.
- *R* is the ideal gas constant.

➤ To derive the Eyring equation, we start with the transition state theory assumptions:

1. The rate of a chemical reaction is determined by the rate at which reactant molecules reach the transition state.
2. The transition state is in dynamic equilibrium with the reactants and has a specific Gibbs free energy of activation ($\Delta G^\ddagger$).

➤ According to statistical mechanics, the rate constant can be expressed using the principles of transition state theory as follows:

$$k = (k_b * T / h) * e^{(-\Delta G^\ddagger / RT)}$$

To understand this equation, let's break it down:

- 1) $(k_b * T / h)$: This term represents the frequency of molecular collisions and their energy distribution. It

accounts for the number of available energy states and the probability of molecules having sufficient energy to overcome the activation barrier.

2) $e^{(-\Delta G^\ddagger/RT)}$: This term incorporates the exponential factor, which relates to the probability of reactant molecules having sufficient energy to surpass the activation energy barrier and form the transition state.

- $\Delta G^\ddagger/RT$ represents the ratio of the Gibbs free energy of activation to the product of the gas constant (R) and the temperature (T). It measures the energy difference between the reactants and the transition state and reflects the energy required to reach the transition state.
- $e^{(-\Delta G^\ddagger/RT)}$ is the Boltzmann factor that accounts for the probability of molecules having energy greater than or equal to the activation energy.

Combining these terms, we obtain the Eyring equation, which provides a quantitative relationship between the rate constant, temperature, activation energy, and other fundamental constants.

It's important to note that the Eyring equation assumes the reaction follows the principles of transition state theory and requires knowledge of the Gibbs free energy of activation. Additionally, the equation assumes the reaction takes place in the absence of other complex factors such as diffusion, catalysis, or surface effects.

